

고급 프로그래밍_인트로

1. 강의 계획서
2. 파이썬 프로그램 UI 설치 및 통일
3. 예비 실습 (계산기 실행파일)

1. 강의 계획서

수업일정표 입력

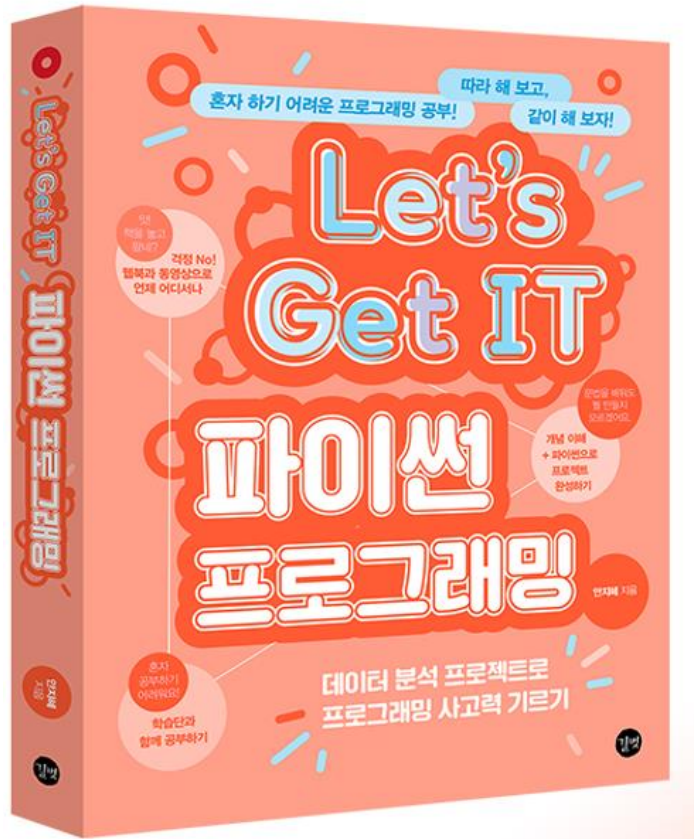
저장

교과목명		고급프로그래밍2	교과목 코드	14107001	이수구분	전선	학점	3
주차	기간	수업내용 및 학습활동					비고	
1	09/01 ~ 09/07	파이썬 프로그램 리뷰 (1)						
2	09/08 ~ 09/14	파이썬 프로그램 리뷰 (2)						
3	09/15 ~ 09/21	Case Study I						
4	09/22 ~ 09/28	Case Study II						
5	09/29 ~ 10/05	Project I						
6	10/06 ~ 10/12	Project I						
7	10/13 ~ 10/19	Project I 발표						
8	10/20 ~ 10/26	Project II						
9	10/27 ~ 11/02	Project II						
10	11/03 ~ 11/09	Project II 발표						
11	11/10 ~ 11/16	Project III						
12	11/17 ~ 11/23	Project III						
13	11/24 ~ 11/30	Project III						
14	12/01 ~ 12/07	Project III 발표						
15	12/08 ~ 12/14	서면 고사						

과제	성적평가 및 방법	강의개요 및 목표	전년도 대비 수업 개선사항	핵심역량 및 강의진행방법	강의진행 및 학습자 유의사항	교재 및 참고자료	수업 만족도
평가요소	평가방법						비율(%)
출석	5회 결석 시 F처리						20
중간고사	필기시험						20
기말고사	프로젝트 시험						30
레포트	프로젝트 발표 2회						30

1. 강의 계획서

주 강의 자료



클래스

14.1.1 클래스와 객체

클래스와 객체의 관계

- 클래스(class) : 캐릭터들이 가져야 할 공통 특성과 기능을 모아 놓은 것, 모든 캐릭터가 가져야 하는 형식을 정의한 것
- 객체(object) : 클래스를 바탕으로 만들어진 실체, 인스턴스(instance)라고도 함

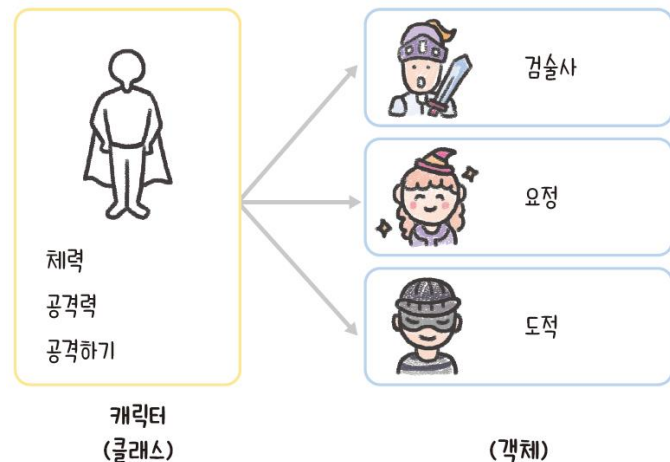
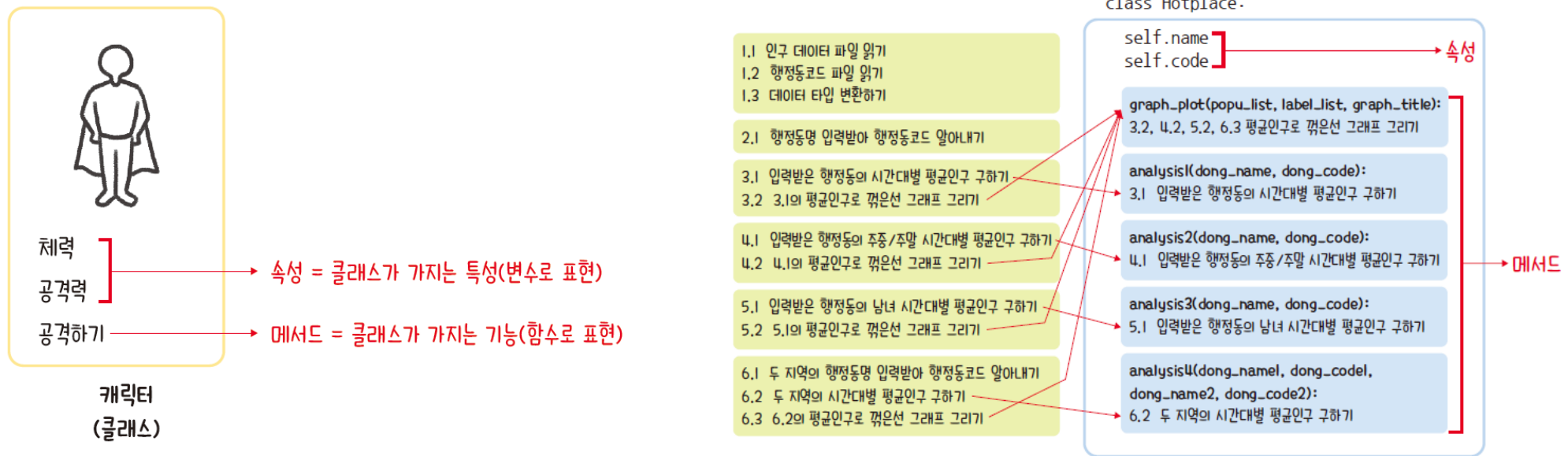


그림 14-1 클래스와 객체의 관계

클래스

속성과 메서드

- 속성(attribute) : 클래스의 특성을 저장하는 변수
- 메서드(method) : 클래스의 기능을 표현하는 사용자 정의 함수



```

import matplotlib.pyplot as plt
import datetime

class Hotplace :
    def __init__(self, dong_name, dong_code) :
        self.name = dong_name
        self.code = dong_code

    def graph_plot(self, popu_list, label_list, graph_title) :
        plt.rc('font', family='Malgun Gothic')
        plt.title(graph_title)
        for i in range(len(popu_list)) :
            plt.plot(range(24), popu_list[i], label = label_list[i])
        plt.legend()
        plt.xlabel('시간대')
        plt.ylabel('평균인구수')
        plt.xticks(range(24), range(24))
        plt.show()

# analysis4()에서 다른 지역을 넣어 호출하는 부분이 있어 매개변수를 두어야 함
def analysis1(self, dong_name, dong_code, graph_yn = 'y') :
    population = [0 for i in range(24)]
    for row in data :
        if row[2] == dong_code :
            time, p = row[1], row[3]
            population[time] += p
    population = [p/31 for p in population]

    if graph_yn == 'n' :
        return population

    population = [population]
    labels = ['평균인구']
    title = dong_name + ' 시간대별 평균인구 그래프'
    self.graph_plot(popu_list= population, label_list= labels, graph_title= title)

```

```

def analysis2(self) :
    weekday = [0 for i in range(24)]
    weekend = [0 for i in range(24)]

    for row in data :
        if row[2] == self.code :
            time, p = row[1], row[3]
            year, mon, day = int(row[0][:4]), int(row[0][4:6]), int(row[0][6:])
            num = datetime.date(year, mon, day).weekday()
            if num < 5 :
                weekday[time] += p
            else :
                weekend[time] += p
    weekday = [w/22 for w in weekday]
    weekend = [w/9 for w in weekend]

    data_set = [weekday, weekend]
    labels = ['평일', '주말']
    title = self.name + ' 평일/주말 시간대별 인구'
    self.graph_plot(popu_list= data_set, label_list= labels, graph_title= title)

def analysis3(self) :
    male = [0 for i in range(24)]
    female = [0 for i in range(24)]

    for row in data :
        if row[2] == self.code :
            time = int(row[1])
            male[time] += sum(row[4:18]) # 남성 인구 총합 구하기
            female[time] += sum(row[18:32]) # 여성 인구 총합 구하기

    male = [m/31 for m in male]
    female = [f/31 for f in female]

    data_set = [male, female]
    labels = ['남성', '여성']
    title = self.name + ' 남녀 시간대별 인구 그래프'
    self.graph_plot(popu_list= data_set, label_list= labels, graph_title= title)

def analysis4(self, dong_name2, dong_code2) :
    population1 = self.analysis1(dong_name=self.name, dong_code=self.code, graph_yn='n')
    population2 = self.analysis1(dong_name=dong_name2, dong_code=dong_code2, graph_yn='n')

    data_set = [population1, population2]
    labels = [self.name, dong_name2]
    title = self.name + '/' + dong_name2 + ' 시간대별 그래프'
    self.graph_plot(popu_list= data_set, label_list= labels, graph_title= title)

```

```
import csv
```

```
def file_open() :
    f = open('LOCAL_PEOPLE_DONG_201912.csv', encoding = 'utf8')
    data = csv.reader(f)
    next(data)
    data = list(data)
```

```
    f2 = open('dong_code.csv', encoding = 'utf8')
    code_data = csv.reader(f2)
    next(code_data)
    next(code_data)
    code_data = list(code_data)
```

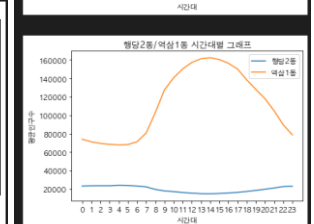
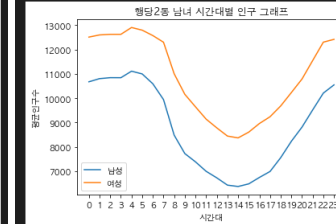
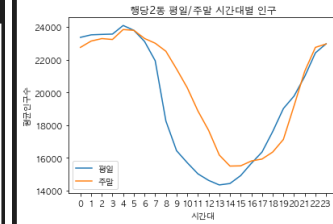
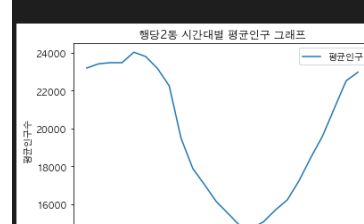
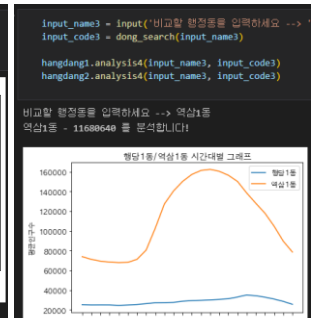
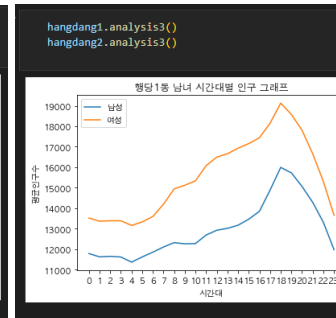
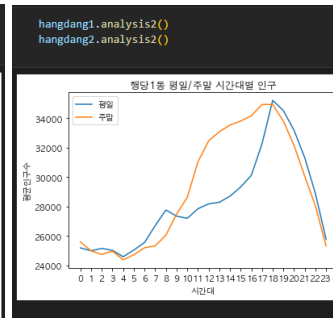
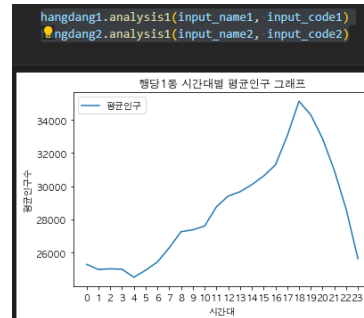
```
    for row in data :
        for i in range(1, 32) :
            if i <= 2 :
                row[i] = int(row[i])
            else :
                row[i] = float(row[i])
    for row in code_data :
        row[1] = int(row[1])
```

```
    return data, code_data
```

```
def dong_search(dong_name) :
    for row in code_data :
        if row[-1] == dong_name :
            code = row[1]
    print(dong_name, '-', code, '를 분석합니다!')
    return code
```

```
data, code_data = file_open()
input_name1 = input('핫플레이스가 위치한 행정동을 입력하세요 --> ')
input_code1 = dong_search(input_name1)
hangdang1 = Hotplace(input_name1, input_code1)
input_name2 = input('핫플레이스가 위치한 행정동을 입력하세요 --> ')
input_code2 = dong_search(input_name2)
hangdang2 = Hotplace(input_name2, input_code2)
```

행당1동



참고자료

2 파이썬 프로그램 설치 및 통일


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2 파이썬 프로그램 네 설치 및 통일

파이썬 코딩 환경 설명

윈도의 <시작> 버튼

→ [모든 프로그램]-[Python 3.6]-[IDLE (Python 3.6 32-bit)] 메뉴 선택

→ IDLE이 시작되며 파이썬 셸(Python Shell)이 대화형 모드로 나타남

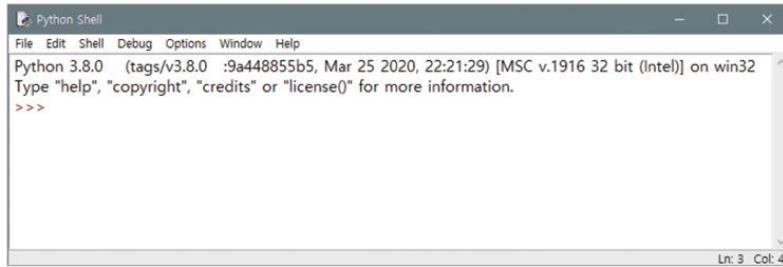


그림 1-11 IDLE 실행 화면

예1

>>> 다음에 `print("Hello, world!")`를 입력하고 [Enter]

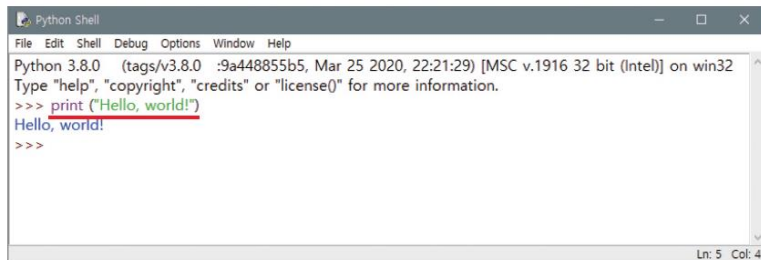


그림 1-12 코드 입력과 실행 예 1

예2

>>> 다음에 다음 계산식을 입력하고 [Enter]

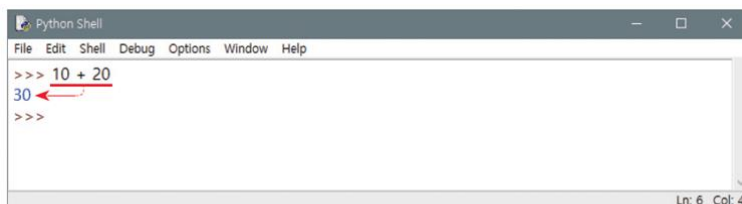


그림 1-13 코드 입력과 실행 예 2

파이썬 IDLE 종료

[File]-[Exit] 메뉴 선택

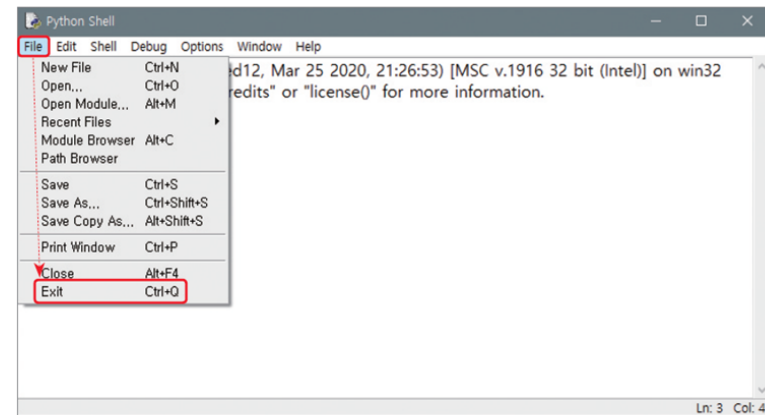


그림 1-15 IDLE 종료

2 파이썬 프로그램 N 설치 및 통일

Vs code 설치 <https://code.visualstudio.com/>

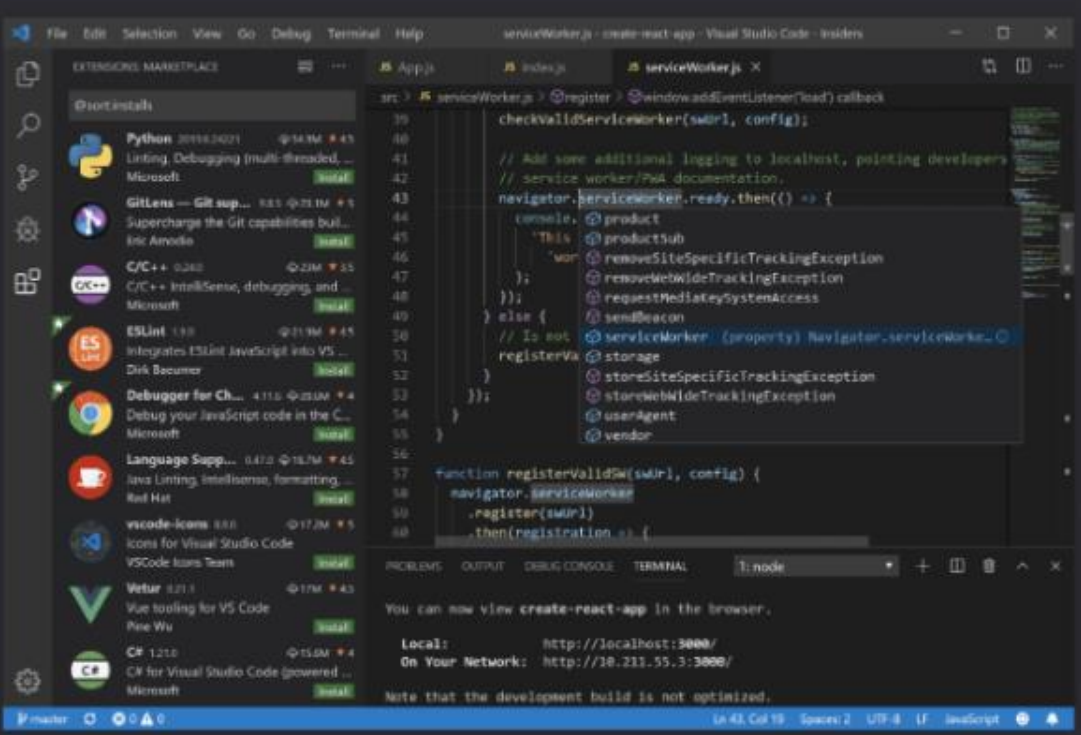
Code editing.
Redefined.

Free. Built on open source. Runs everywhere.

Download for Windows
Stable Build

Other platforms and Insiders Edition

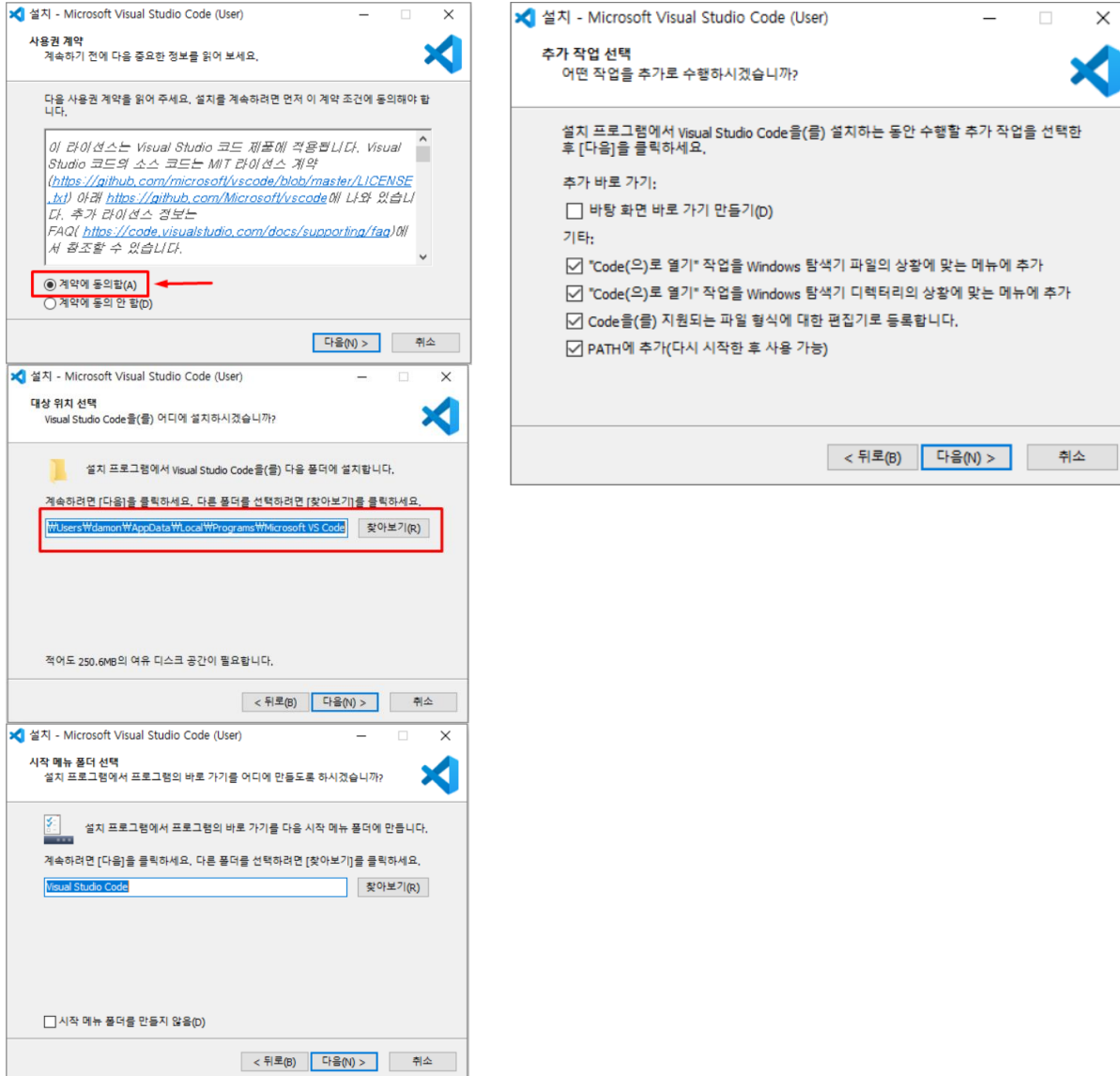
By using VS Code, you agree to its [license and privacy statement](#).



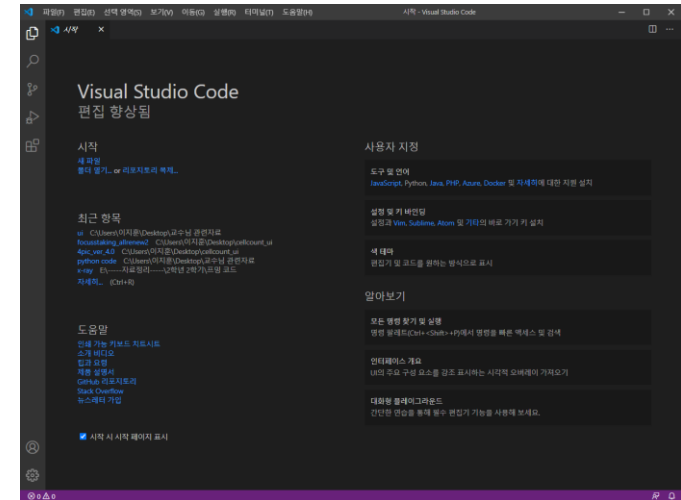
The screenshot shows the Visual Studio Code interface. On the left is the 'EXTENSIONS MARKETPLACE' sidebar with a list of installed and available extensions like Python, GitLens, C/C++, ESLint, and others. The main editor area shows a JavaScript file named 'serviceWorker.js' with code for registering a service worker. The bottom panel shows the 'TERMINAL' with the output of a 'create-react-app' command, indicating that the application is now viewable in the browser at 'http://localhost:3000/'.

2 파이썬 프로그램 N 설치 및 통일

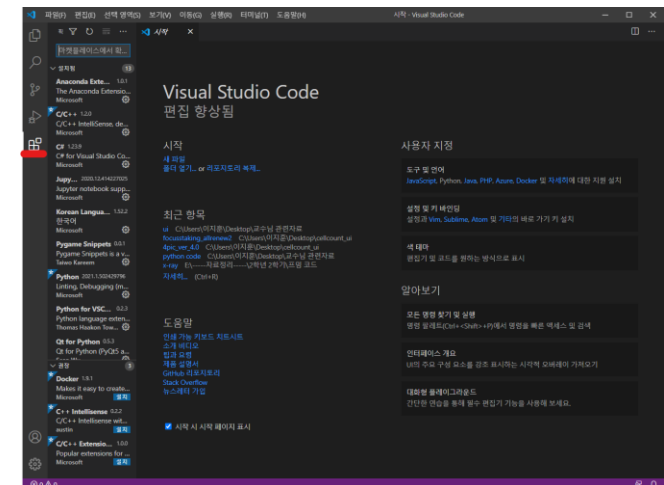
1. 링크 접속 후 다운로드 실행



2. PATH에 추가 체크



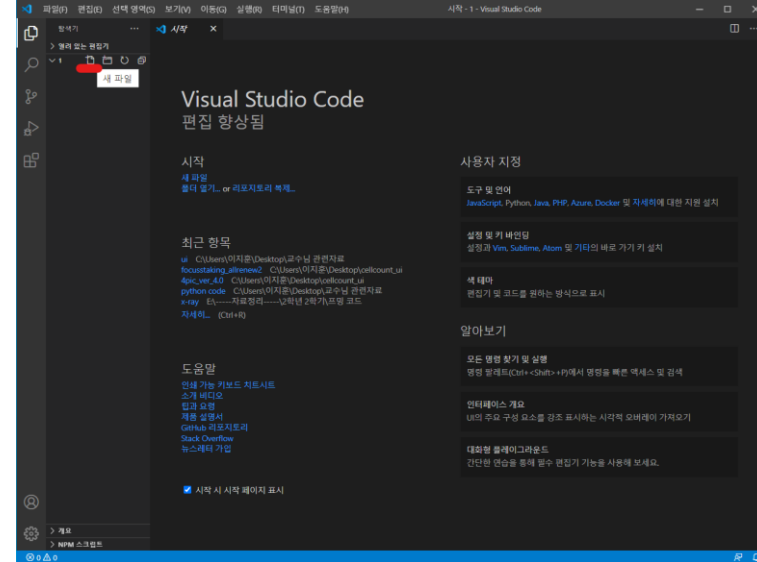
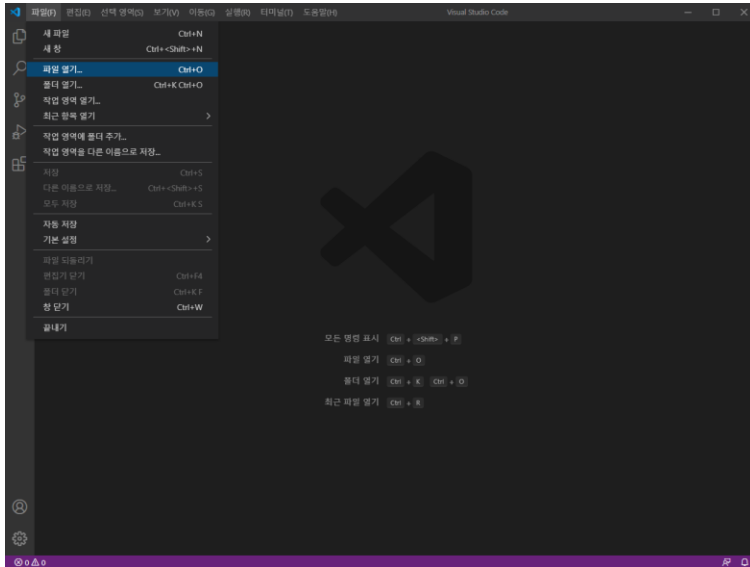
3. 실행



2 파이썬 프로그램 N 설치 및 통일

4. 파이썬 연동

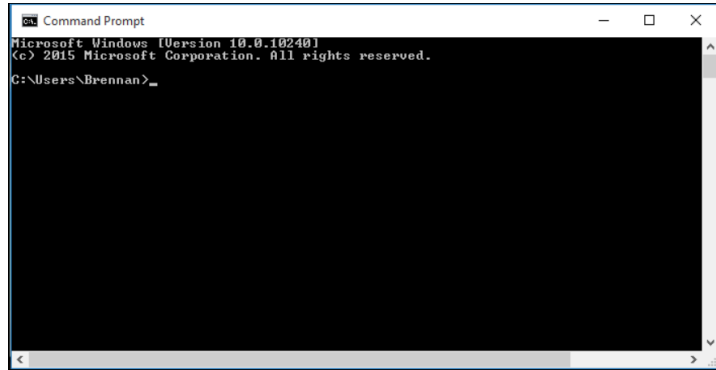
- 'python' 검색 후 설치
- 'Korean Language Pack for Visual Studio Code' 후 Visual Studio Code 재시작



- ### 5. 폴더 열기로 원하는 위치의 폴더를 열어서 코드 작성
- 파일명은 항상 .py

[참고자료] PYQT5 설치

UI를 제작하기 위해 pyqt5를 설치한다.



[cmd(명령 프롬프트)]
pip install pyqt5 입력

pip install pyqt5-tools

```

pip install pyqt5-tools
Collecting pyqt5-tools
  Downloading pyqt5_tools-5.13.2.1.6rc1-cp38-none-win32.whl (56.9 MB)
    56.9 MB 6.8 MB/s
Collecting click
  Downloading click-7.1.2-py2.py3-none-any.whl (82 kB)
    82 kB 483 kB/s
Collecting pyqt5==5.13.2
  Downloading PyQt5-5.13.2-5.13.2-cp35.cp36.cp37.cp38-none-win32.whl (43.6 MB)
    43.6 MB 6.8 MB/s
Collecting python-dotenv
  Downloading python_dotenv-0.13.0-py2.py3-none-any.whl (17 kB)
Requirement already satisfied: PyQt5_sip<13,>=4.19.19 in d:\blogproject\venv\lib\site-packages (from pyqt5==5.13.2->pyqt5-tools) (12.7.2)
Installing collected packages: click, pyqt5, python-dotenv, pyqt5-tools
Attempting uninstall: pyqt5
  Found existing installation: PyQt5 5.14.2
  Uninstalling PyQt5-5.14.2:
    Successfully uninstalled PyQt5-5.14.2
Successfully installed click-7.1.2 pyqt5-5.13.2 pyqt5-tools-5.13.2.1.6rc1 python-dotenv-0.13.0
  
```

designer.exe 위치

- ...Python\Python버전\Lib\site-packages\pyqt5_tools\Qt\bin

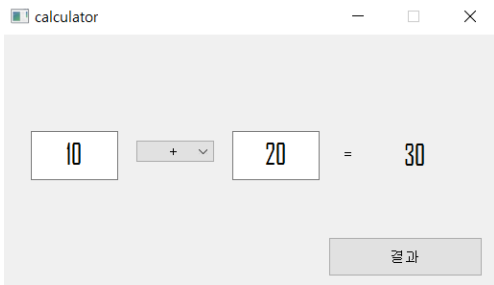
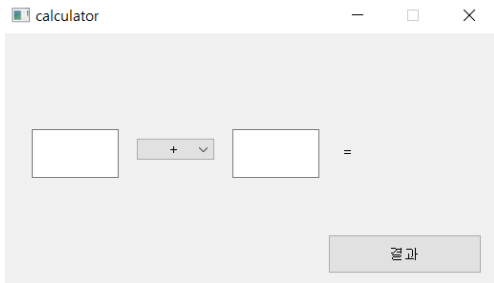


designer.exe 실행

```

pip install pyqt5
Collecting pyqt5
  Downloading PyQt5-5.14.2-5.14.2-cp35.cp36.cp37.cp38-none-win32.whl (44.8 MB)
    44.8 MB 726 kB/s
Collecting PyQt5-sip<13,>=12.7
  Downloading PyQt5_sip-12.7.2-cp38-cp38-win32.whl (49 kB)
    49 kB ...
Installing collected packages: PyQt5-sip, pyqt5
Successfully installed PyQt5-sip-12.7.2 pyqt5-5.14.2
  
```

간단한 계산기



📖 간단한 계산기 만들기

📖 if 문을 이용한 사칙연산

- 상황에 따른 계산법 익히기

if sign == '+':

fin = num1 + num2

elif sign == '-':

fin = num1 - num2

elif sign == '/':

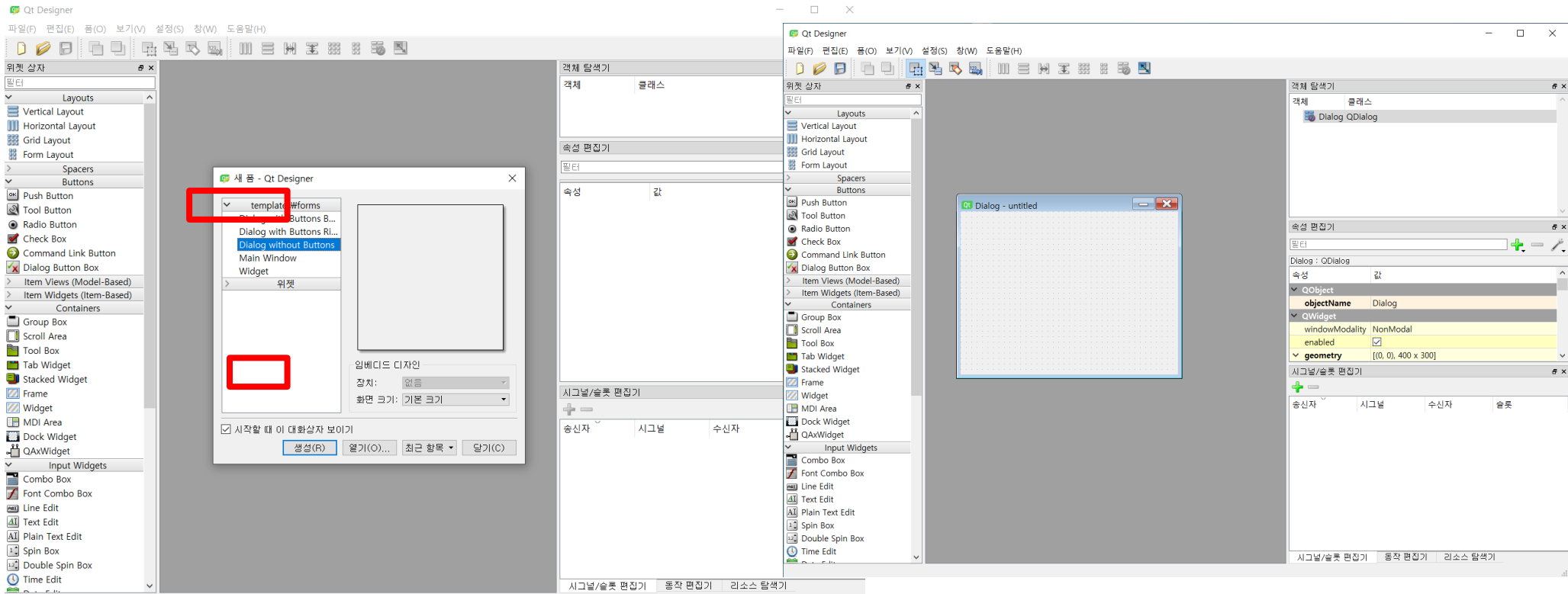
fin = num1 / num2

elif sign == '*':

fin = num1 * num2

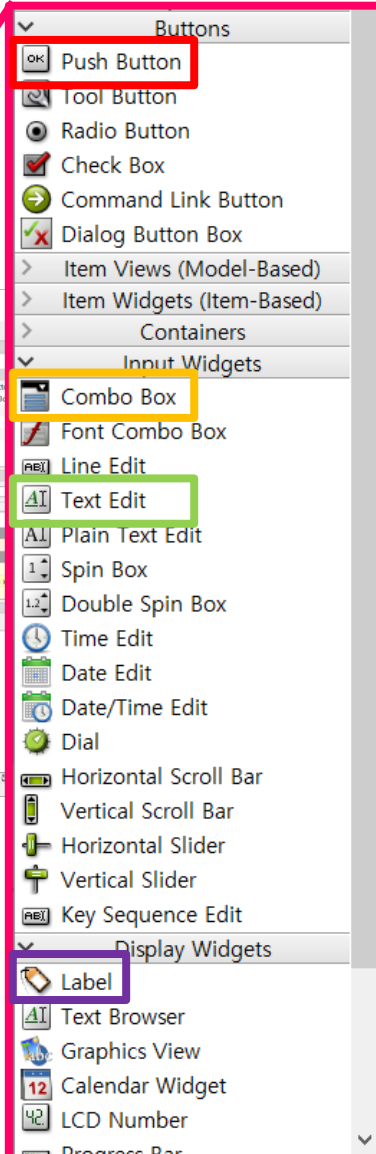
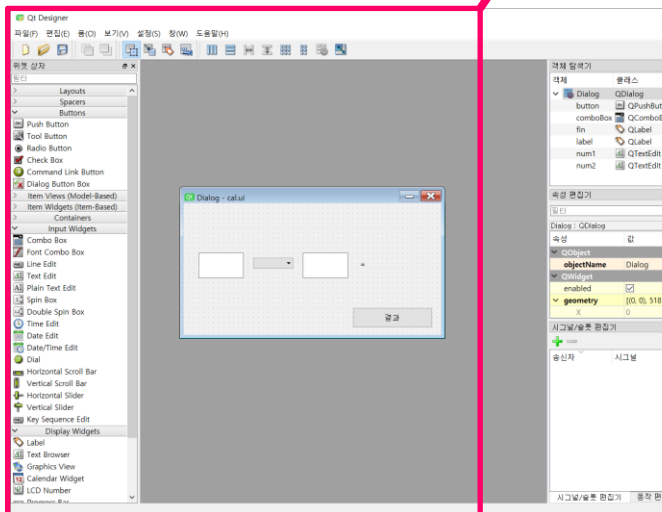
UI 제작

간단한 계산기

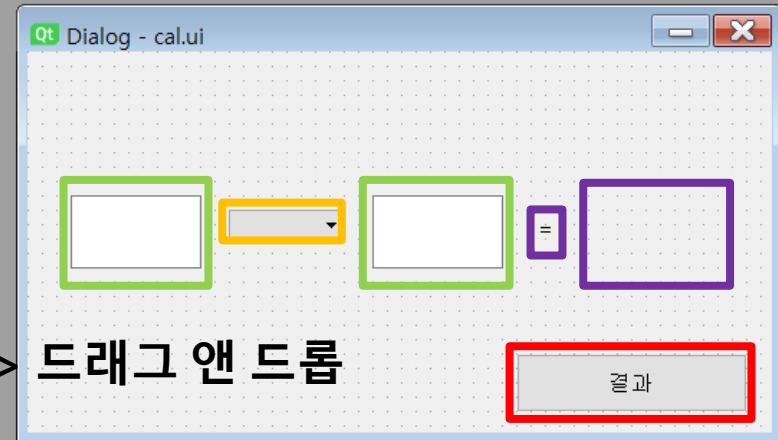


UI 제작

간단한 계산기

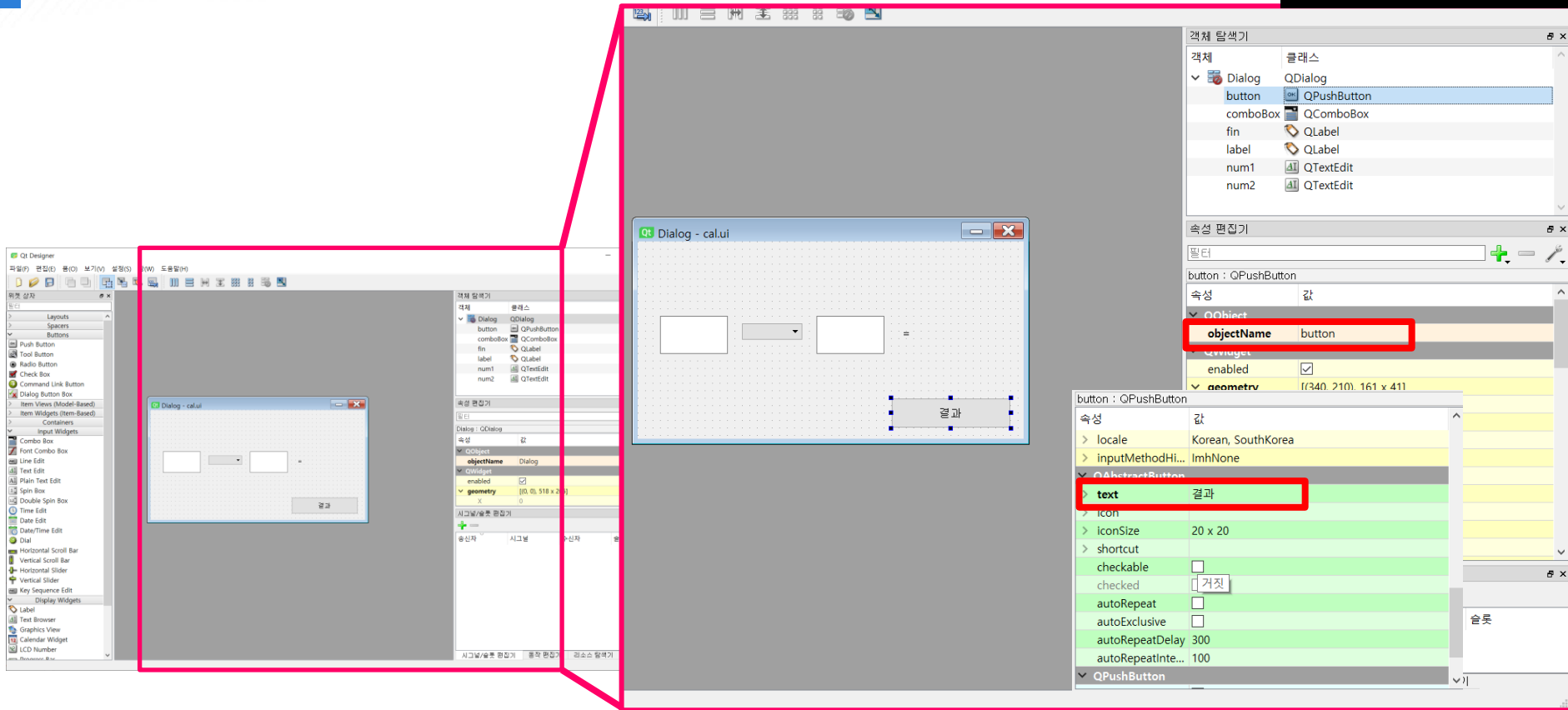


>>> 드래그 앤 드롭



UI 제작

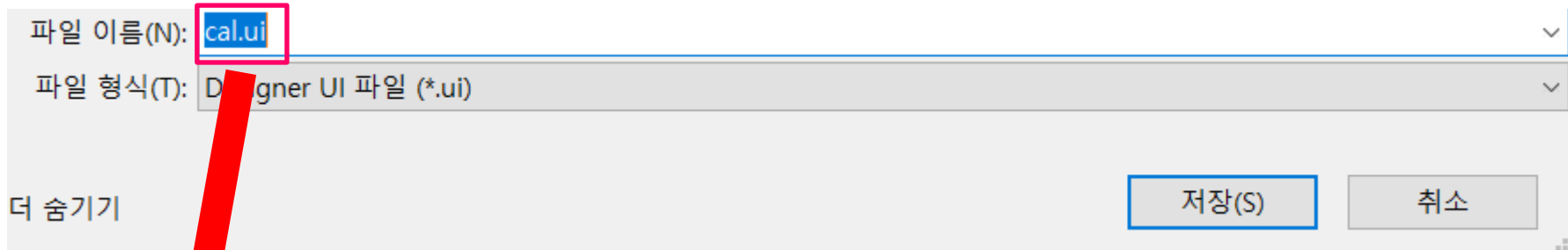
간단한 계산기



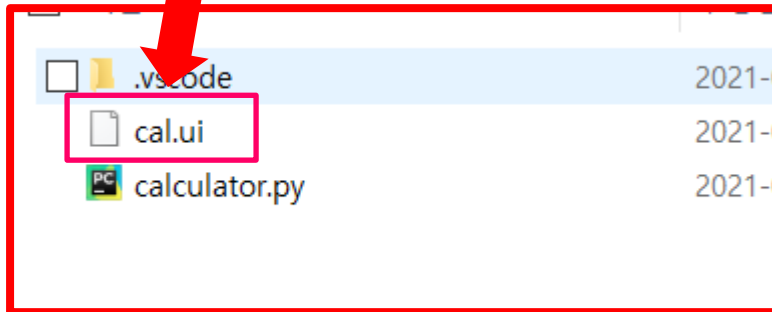
클래스	objectName	Text
QPushButton	Resultl_button	결과
QComboBox	comboBox	
QLabel	fin / label	Label의 텍스트(=)
QTextEdit	num1 / num2	

UI 제작

간단한 계산기

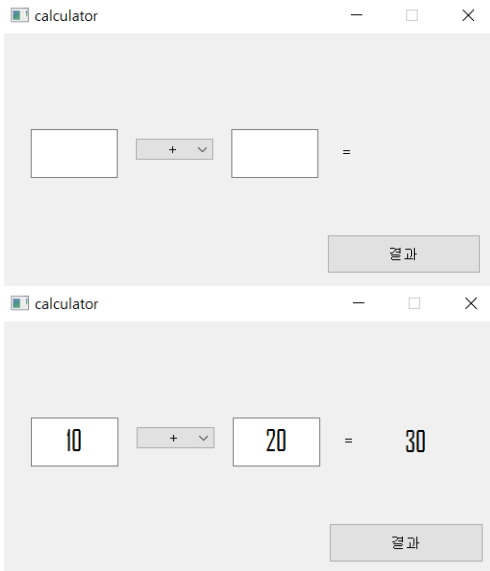


>>> ui를 실행 py와 같은 폴더에 저장



Calculator.py

간단한 계산기



```
import sys
import math
from PyQt5.QtCore import *
from PyQt5.QtGui import *
from PyQt5.QtWidgets import *
from PyQt5 import uic
```

```
ui_form = uic.loadUiType( "cal.ui" )[0]
```

```
class QtCalculator(QMainWindow, ui_form):
    def __init__( self ):
        super(QtCalculator, self).__init__()
        self.setupUi( self )
```

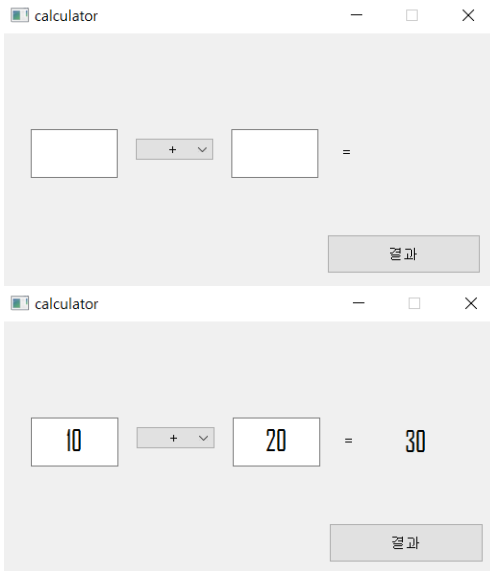
```
if __name__ == "__main__":
    app = QApplication(sys.argv)
    myWindow = QtCalculator()
    myWindow.show()
    app.exec_()
```



>>> ui 불러오기

Calculator.py

간단한 계산기



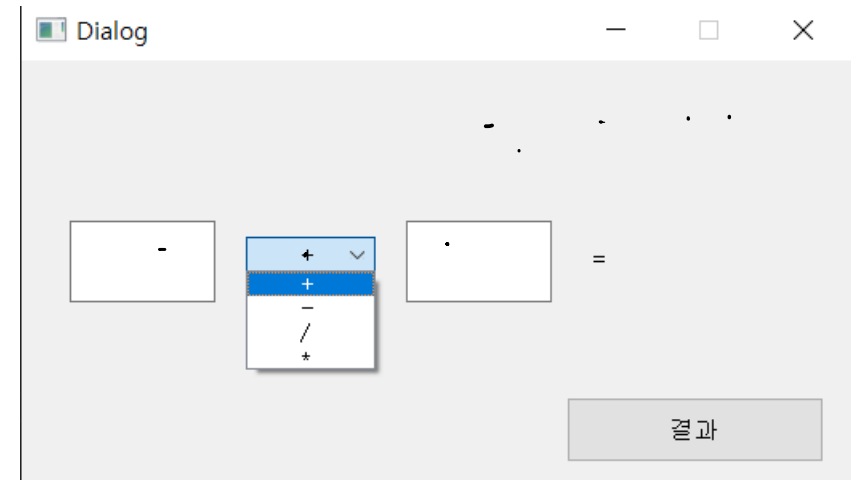
```
import sys
import math
from PyQt5.QtCore import *
from PyQt5.QtGui import *
from PyQt5.QtWidgets import *
from PyQt5 import uic
```

```
ui_form = uic.loadUiType( "cal.ui" )[0]
```

```
class QtCalculator(QMainWindow, ui_form):
    def __init__( self ):
        super(QtCalculator, self).__init__()
        self.setupUi( self )
```

```
self.comboBox.addItem(' +')
self.comboBox.addItem(' -')
self.comboBox.addItem(' /')
self.comboBox.addItem(' *')
```

```
if __name__ == "__main__":
    app = QApplication(sys.argv)
    myWindow = QtCalculator()
    myWindow.show()
    app.exec_()
```



>>> comboBox에 기호 채우기

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```
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class QtCalculator(QMainWindow,
ui_form):
    def __init__( self ):
        super(QtCalculator, self ).__i
nit__()
        self.setupUi( self )
```

```
        self.comboBox.addItem('
+')
        self.comboBox.addItem('
')
        self.comboBox.addItem('
/')
        self.comboBox.addItem('
*')
```

```
        self.button.clicked.connect(se
lf.button_click)
```

```
def button_click(self):
    try:
```

```
        num1 = int(self.num1.toPla
inText())
        num2 = int(self.num2.toPla
inText())
        sign = self.comboBox.curr
entText()
        if sign == ' + ':
```

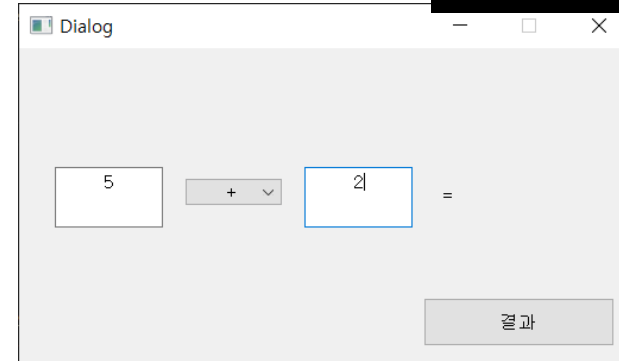
```
            fin = num1 + num2
        elif sign == ' - ':
            fin = num1 - num2
        elif sign == ' / ':
            fin = num1 / num2
        elif sign == ' * ':
            fin = num1 * num2
```

```
        self.fin.setText(str(fin))
```

```
    except:
```

```
        pass
```

```
if __name__ == "__main__":
    app = QApplication(sys.argv)
    myWindow = QtCalculator()
    myWindow.show()
    app.exec_()
```



>>> int(self.num1.toPlainText())
Text 읽어들이기

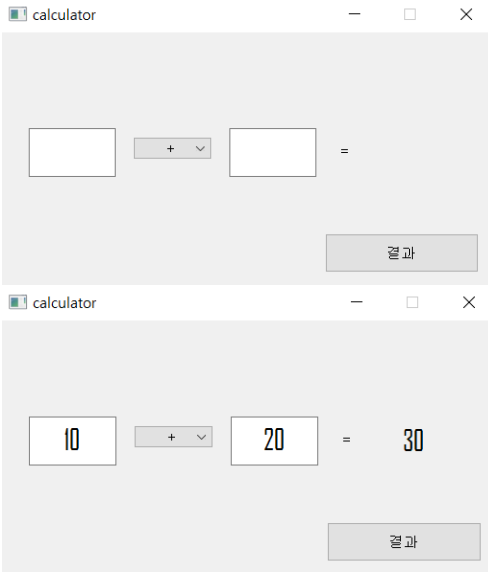
>>> 입력된 숫자와 기호를 통해 계산



>>> 결과값 Text에 도출

Calculator.py

간단한 계산기



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from PyQt5.QtCore import *
from PyQt5.QtGui import *
from PyQt5.QtWidgets import *
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ui_form = uic.loadUiType( "cal.ui" )[0]
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class QtCalculator(QMainWindow, ui_form):
    def __init__( self ):
        super(QtCalculator, self ).__init__()
        self.setupUi( self )
```

```
self.comboBox.addItem(' +')
self.comboBox.addItem(' -')
self.comboBox.addItem(' /')
self.comboBox.addItem(' *')
```

```
self.button.clicked.connect(self.button_click)
```

```
def button_click(self):
    try:
        num1 = int(self.num1.toPlainText())
        num2 = int(self.num2.toPlainText())
        sign = self.comboBox.currentText()
        if sign == ' +':
            fin = num1 + num2
        elif sign == ' -':
            fin = num1-num2
        elif sign == ' /':
            fin = num1 / num2
```

```
        elif sign == ' *':
            fin = num1 * num2

        self.fin.setText(str(fin))
    except:
        pass
```

```
if __name__ == "__main__": #메인함수이면 실행
    app = QApplication(sys.argv) #PyQt 이벤트 루프 실행
    myWindow = QtCalculator()
    myWindow.show()
    app.exec_()
```

실행파일 만들기

CMD에 pip install pyinstaller 입력

```
C:\> 선택 명령 프롬프트 - pip install pyinstaller

(venv) C:\Users\2018.08.21.01\PycharmProjects\TensorFlow_EX\venv\Scripts>pip install pyinstaller
Collecting pyinstaller
Collecting altgraph (from pyinstaller)
  Using cached https://files.pythonhosted.org/packages/0a/cc/646187eac4b797069e2e6b736f14cdef85dbe405c9bfc7803ef36e4f62ef/altgraph-0.16.1-py2.py3-none-any.whl
Collecting pywin32-ctypes (from pyinstaller)
  Using cached https://files.pythonhosted.org/packages/9e/4b/3ab2720f1fa4b4bc924ef1932b842edf10007e4547ea8157b0b9fc78599a/pywin32-ctypes-0.2.0-py2.py3-none-any.whl
Collecting pefile>=2017.8.1 (from pyinstaller)
Collecting macholib>=1.8 (from pyinstaller)
  Using cached https://files.pythonhosted.org/packages/41/f1/6d23e1c79d68e41eb592338d90a33af813f98f2b04458aaf0b86908da2d8/macholib-1.11-py2.py3-none-any.whl
Requirement already satisfied: setuptools in c:\users\2018.08.21.01\pycharmprojects\tensorflow_ex\venv\lib\site-packages\setuptools-40.8.0-py3.6.egg (from pyinstaller) (40.8.0)
Collecting future (from pefile>=2017.8.1->pyinstaller)
Installing collected packages: altgraph, pywin32-ctypes, future, pefile, macholib, pyinstaller
-
```

실행파일 만들기

UI를 작동하기 위해 UI를 절대 경로로 설정하는 코드

```
import math
import sys
import os

from PyQt5.QtCore import *
from PyQt5.QtGui import *
from PyQt5.QtWidgets import *
from PyQt5 import uic

def resource_path(relative_path):
    base_path = getattr(sys, '_MEIPASS', os.path.dirname(os.path.abspath(__file__)))
    return os.path.join(base_path, relative_path)

form = resource_path("Calculator_UI.ui")
form_class = uic.loadUiType(form)[0]

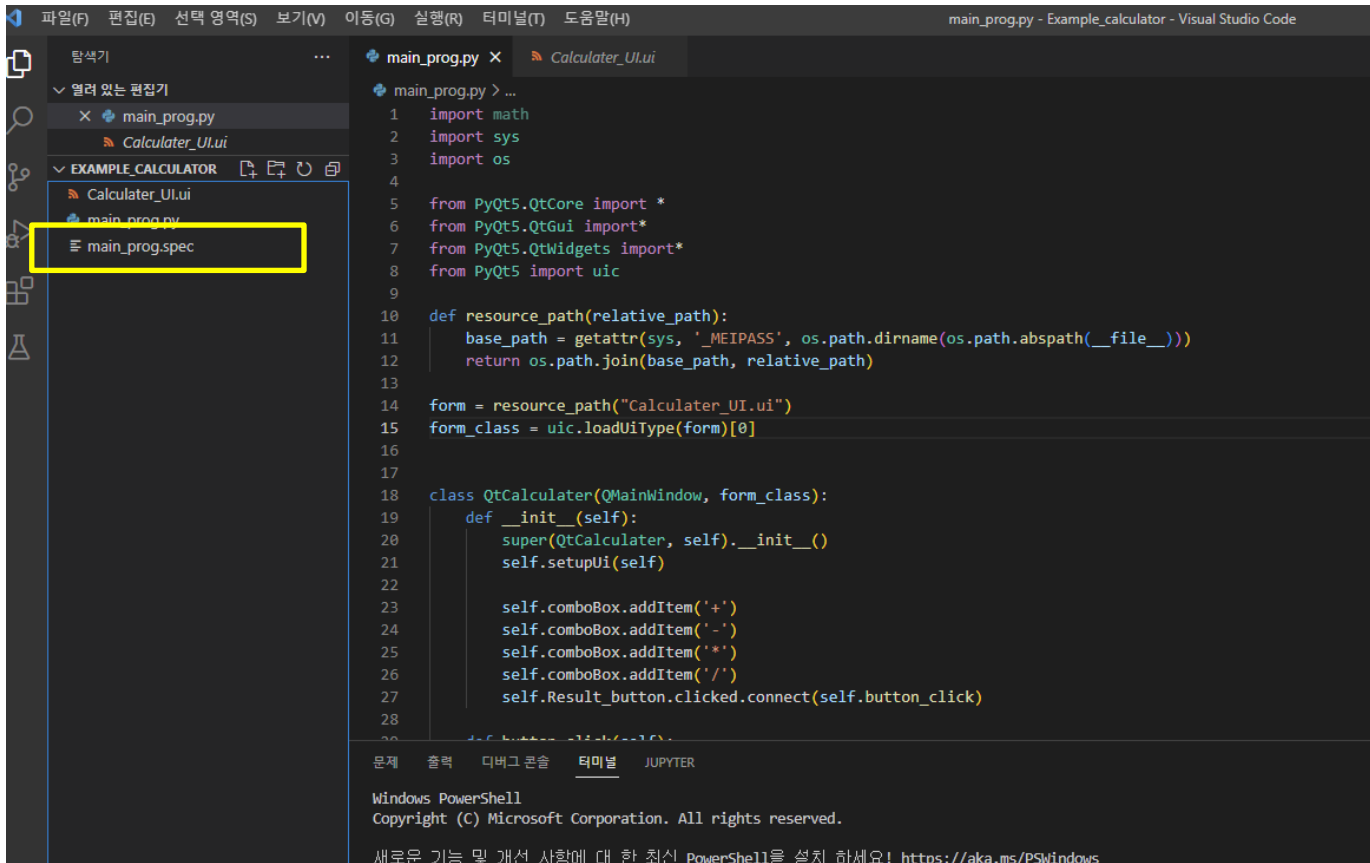
class QtCalculator(QMainWindow, form_class):
```

실행파일 만들기

터미널에
입력

```
r> pyi-makespec --noconsole --onefile main_prog.py
python\main_prog.spec
```

pyi-makespec --noconsole --onefile main_prog.py



Main_prog.spec 파일
생성

실행파일 만들기

```
added_files = [('Calclater_UI.ui', '.')]
```

```

# -*- mode: python ; coding: utf-8 -*-

block_cipher = None
added_files = [('Calclater_UI.ui', '.')]

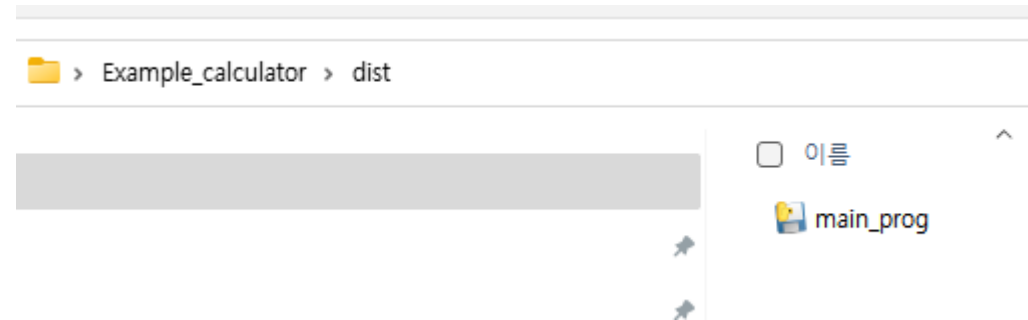
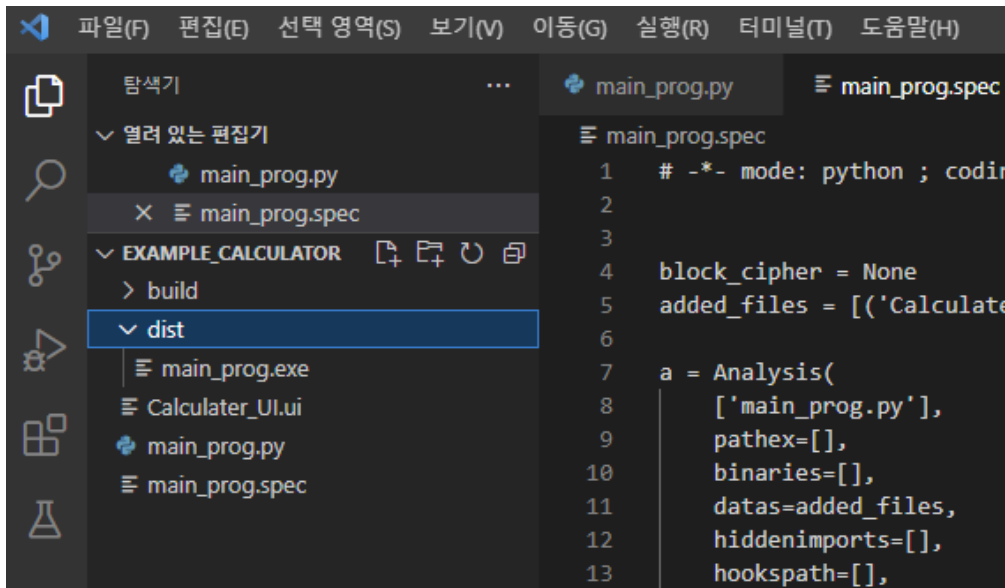
a = Analysis(
    ['main_prog.py'],
    pathex=[],
    binaries=[],
    datas=added_files,
    hiddenimports=[],
    hookspath=[],
    hooksconfig={},
    runtime_hooks=[],
    excludes=[],
    win_no_prefer_redirects=False,
    win_private_assemblies=False,
    cipher=block_cipher
)
  
```

위 두줄 추가후에 저장 후에

터미널에
입력

```
r> pyinstaller --clean main_prog.spec
```

실행파일 만들기



dist 폴더에 실행파일이 생성된것을
확인하과 실행해 본다